



2009

**HIGHER SCHOOL CERTIFICATE
EXAMINATION**

Physics

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
- Draw diagrams using pencil
- Board-approved calculators may be used
- A data sheet, formulae sheets and Periodic Table are provided at the back of this paper
- Write your Centre Number and Student Number at the top of pages 9, 11, 13, 15, 19 and 21

Total marks – 100

Section I Pages 2–22

75 marks

This section has two parts, Part A and Part B

Part A – 15 marks

- Attempt Questions 1–15
- Allow about 30 minutes for this part

Part B – 60 marks

- Attempt Questions 16–27
- Allow about 1 hour and 45 minutes for this part

Section II Pages 23–33

25 marks

- Attempt ONE question from Questions 28–32
- Allow about 45 minutes for this section

Section I

75 marks

Part A – 15 marks

Attempt Questions 1–15

Allow about 30 minutes for this part

Use the multiple-choice answer sheet for Questions 1–15.

- 1 A fast-moving space probe passes close to a planet.

During its journey, how does the gravitational field of the planet affect the speed and direction of the probe?

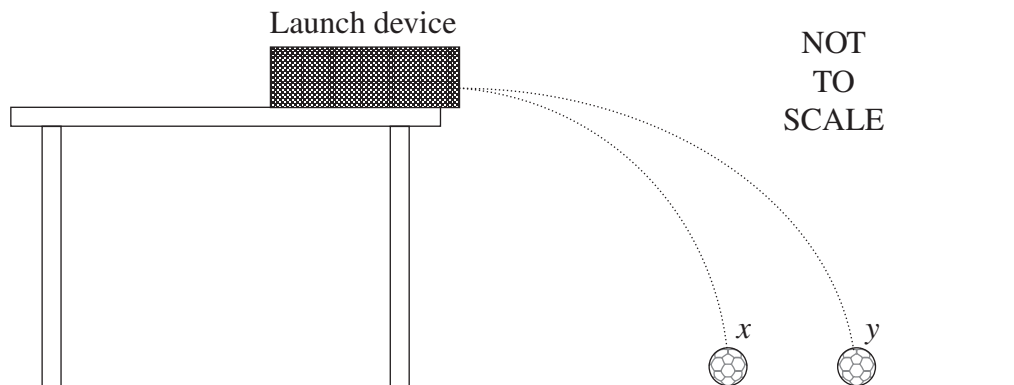
	<i>Speed</i>	<i>Direction</i>
(A)	Remains constant	Remains constant
(B)	Remains constant	Changes
(C)	Changes	Changes
(D)	Changes	Remains constant

- 2 A satellite is moving in a circular orbit of radius 7.0×10^6 m around Earth.

If the speed of the satellite is 8.1×10^3 m s⁻¹, what is its centripetal acceleration?

- (A) 9.4 m s⁻²
(B) 9.8 m s⁻²
(C) 5.6×10^{25} m s⁻²
(D) 3.9×10^{32} m s⁻²
- 3 A satellite is moved from a geostationary orbit to a higher orbit.
- Which statement about the orbit change is correct?
- (A) During the move the gravitational potential energy decreases.
(B) The change in gravitational potential energy is independent of the mass of the satellite.
(C) The work done is the difference between the gravitational potential energy of the higher orbit and that of the geostationary orbit.
(D) The work done is the energy required to move the satellite, which is in the gravitational field, from a very large distance away, to the higher orbit.

- 4 A device launches two identical balls (x and y) simultaneously in a horizontal direction from the same height. The results are shown.



Which statement correctly describes what happens?

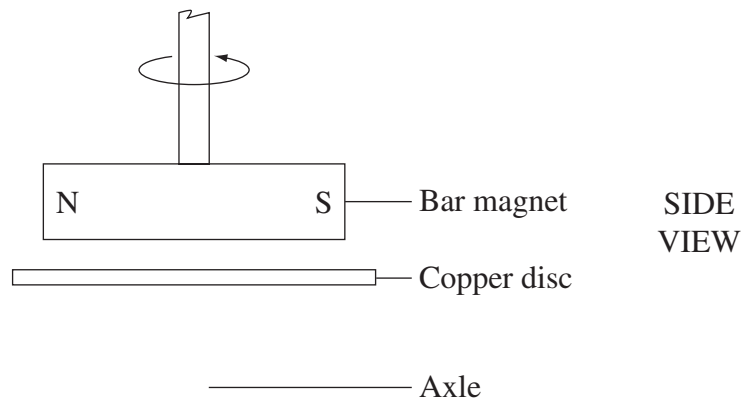
- (A) x hits the ground before y as it is closer to the launch site.
 - (B) y hits the ground before x as it has a higher launch velocity.
 - (C) x and y hit the ground simultaneously with the same velocity.
 - (D) x and y hit the ground simultaneously with different velocities.
- 5 During a lunar eclipse, Earth moves between the Sun and the Moon.



What happens to the force exerted by the Sun on the Moon?

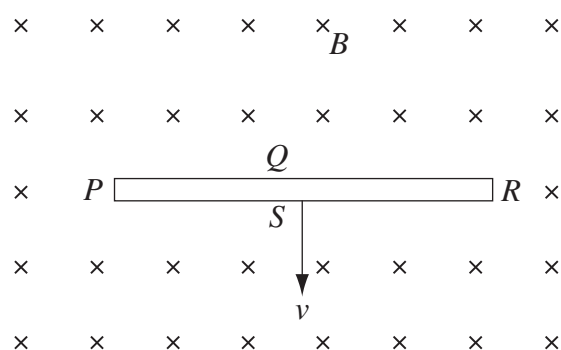
- (A) It increases.
- (B) It decreases.
- (C) It remains unchanged.
- (D) It depends on the closeness of Earth to the Moon.

- 6 Which of the following would increase the output of a simple DC generator?
- (A) Increasing the rotation speed of the rotor
 - (B) Reducing the number of windings in the coil
 - (C) Using slip rings instead of a split ring commutator
 - (D) Wrapping the windings around a laminated, aluminium core
- 7 A type of car speedometer consists of a rotating bar magnet which produces eddy currents in a copper disc. A model of this is shown.



- As the magnet begins to rotate, in which direction does the disc move?
- (A) Toward the magnet
 - (B) Away from the magnet
 - (C) Rotates in the same direction as the magnet
 - (D) Rotates in the opposite direction to the magnet
- 8 What is an essential requirement for the operation of a step-down transformer?
- (A) A laminated iron core
 - (B) A non-conducting core
 - (C) A magnetic interaction between the primary and secondary coils
 - (D) An electrical connection between the primary and secondary coils

- 9 A thin solid conductor with sides $PQRS$ is moving at constant velocity v , at right angles to a uniform magnetic field B , directed into the page as shown.

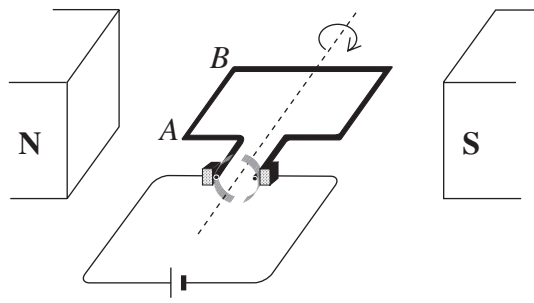


Which side of the conductor has the greatest concentration of electrons?

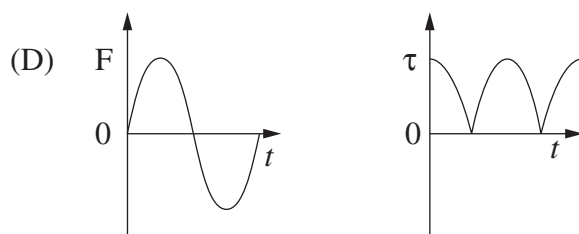
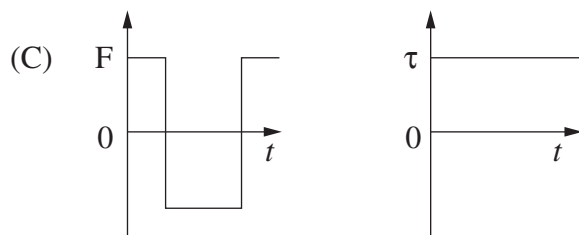
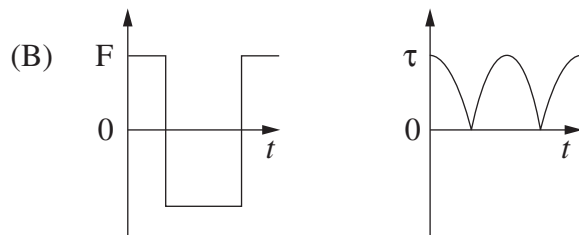
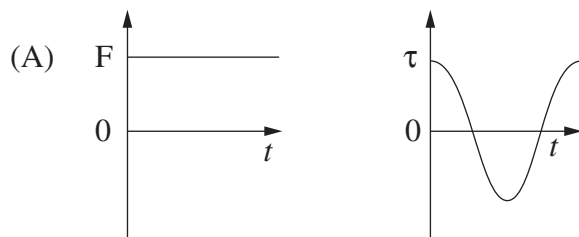
- (A) P
 (B) Q
 (C) R
 (D) S
- 10 Which option best identifies why germanium was replaced by silicon in the semiconductor industry?

	<i>Germanium</i>	<i>Silicon</i>
(A)	Remains a useful semiconductor at higher temperatures	Less abundant
(B)	Less abundant	Remains a useful semiconductor at higher temperatures
(C)	Remains a useful semiconductor at higher temperatures	More abundant
(D)	More abundant	Remains a useful semiconductor at higher temperatures

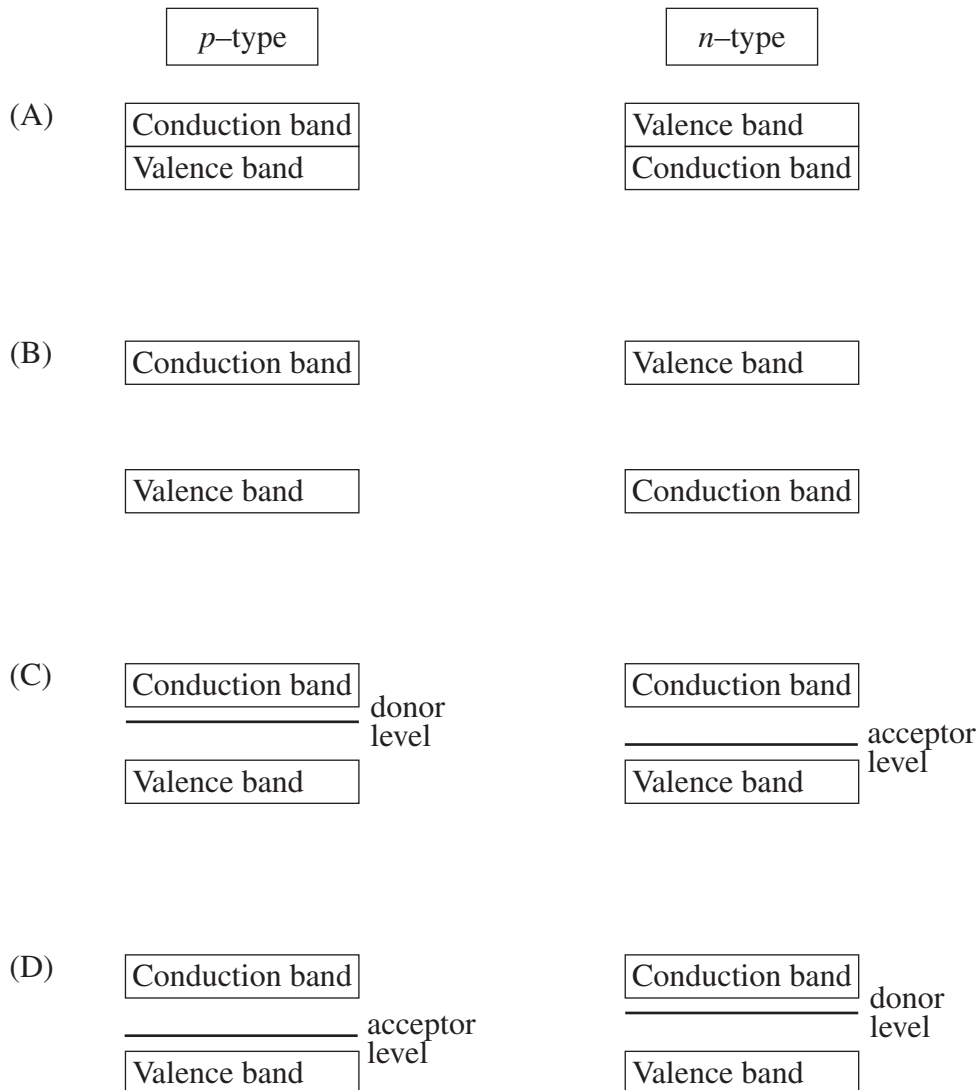
- 11 The diagram shows a DC motor with a constant current flowing to the rotor.



Which pair of graphs best describes the behaviour of the force F on wire AB , and the torque τ on the rotor as functions of time t ?



- 12** Which of the following diagrams best represents the energy bands in p -type and n -type semiconductors?



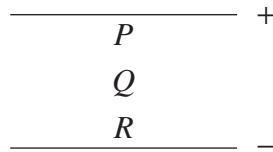
- 13** Why does superconductivity occur in certain materials at low temperatures?

- (A) At low temperatures there are no lattice vibrations.
- (B) Some pairs of electrons experience net attraction at low temperatures.
- (C) The materials are alloys and alloys lose all resistance at low temperatures.
- (D) At low temperatures the materials become magnetic and this reduces the scattering of electron pairs.

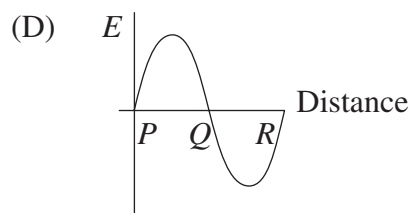
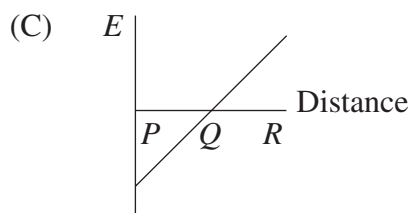
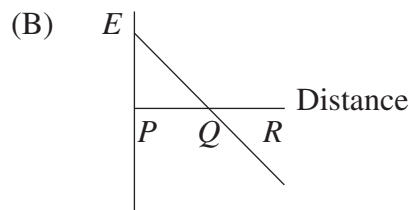
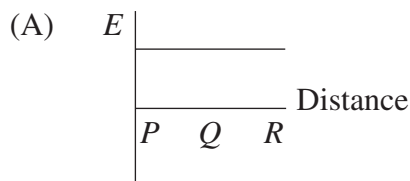
- 14 Blue light is found to cause photoelectric emission from a sodium surface but not from a platinum surface.

Which of the following best accounts for this difference?

- (A) Platinum does not absorb photons.
(B) Platinum has more electrons than sodium.
(C) More energy is needed to remove an electron from a platinum surface.
(D) The intensity of the blue light is not high enough to remove electrons from the platinum surface.
- 15 The diagram shows two parallel plates with opposite charges. P , Q and R represent distances from the positive plate.



Which of the following graphs describes the electric field strength, E , between the plates?



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Centre Number

Section I (continued)

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Student Number

Part B – 60 marks

Attempt Questions 16–27

Allow about 1 hour and 45 minutes for this part

Answer the questions in the spaces provided.

Show all relevant working in questions involving calculations.

Question 16 (3 marks)

NASA recently landed a space probe on an asteroid found between the orbits of Earth and Mars. The 500 kg space probe had a weight of 2.5 N when it landed on the asteroid.

- (a) What would be the weight of this space probe on the surface of Earth? 1

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- (b) Before landing on the asteroid, the space probe was placed in an orbit with radius 50 km. The orbital period was 5.9×10^4 s. 2

What was the mass of the asteroid?

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Question 17 (5 marks)

- (a) Using labelled diagrams, show how a first-hand investigation could be performed to distinguish between an inertial and a non-inertial frame of reference. **2**

- (b) Explain how inertial and non-inertial frames of reference relate to the principle of relativity. **3**

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Centre Number

Section I – Part B (continued)

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Question 18 (4 marks)

The nearest galaxy to ours is the Large Magellanic Cloud, with its centre located 1.70×10^5 light years from Earth. Assume you are in a spacecraft travelling at a speed of $0.99999c$ toward the Large Magellanic Cloud.

- (a) In your frame of reference, what is the distance between Earth and the Large Magellanic Cloud? 2

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- (b) In your frame of reference, how long will it take you to travel from Earth to the Large Magellanic Cloud? 2

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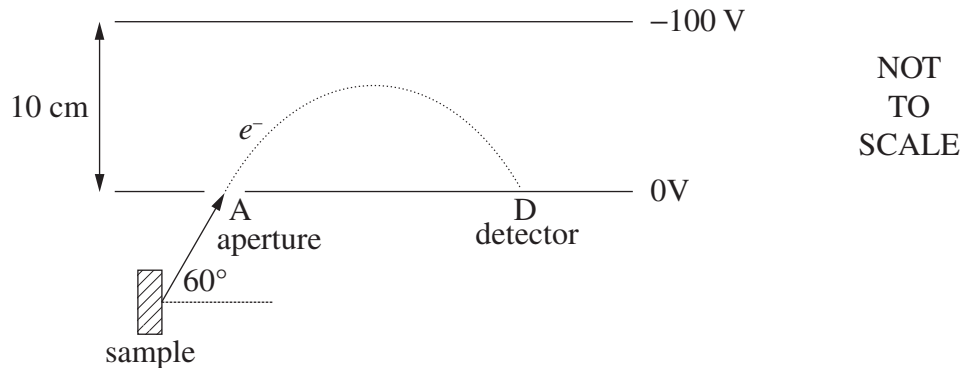
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Question 19 (6 marks)

An electron is emitted from a mineral sample, and travels through aperture A into a spectrometer at an angle of 60° with a speed of $6.0 \times 10^6 \text{ m s}^{-1}$.



- (a) Calculate the magnitude and direction of the force experienced by the electron inside the spectrometer. **3**

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- (b) The electron experiences constant acceleration and eventually strikes the detector, D. **3**

What is the time taken for the electron to travel from A to D?

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Centre Number

Section I – Part B (continued)

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Question 20 (4 marks)

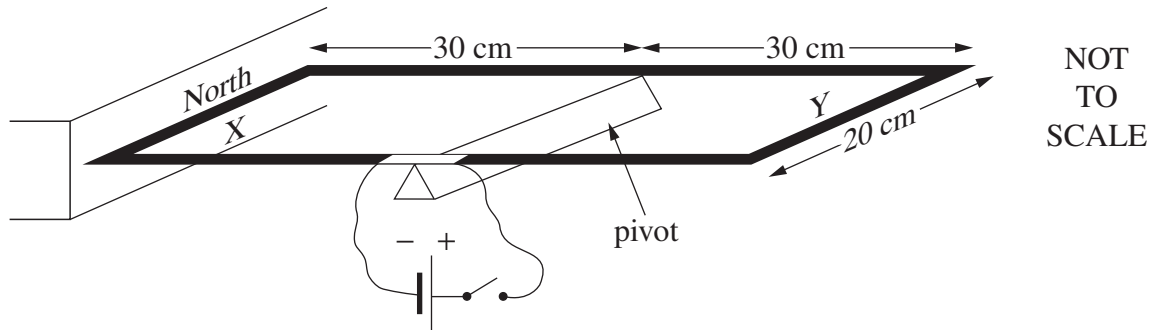
Draw a table to summarise the energy transformations and transfers for three household appliances. Each appliance must have a different type of useful energy output. Include the name of the appliance, its use and the transformation/transfer of energy involved.

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Question 21 (6 marks)

A rectangular wire loop is connected to a DC power supply. Side X of the loop is placed next to a magnet. The loop is free to rotate about a pivot.



When the power is switched on, a current of 20 A is supplied to the loop. To prevent rotation, a mass of 40 g can be attached to either side X or side Y of the loop.

- (a) On which side of the loop should the mass be attached to prevent rotation? **1**

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- (b) Calculate the torque provided by the 40 g mass. **2**

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- (c) Calculate the magnetic field strength around side X. **3**

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Centre Number

Section I – Part B (continued)

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Question 22 (4 marks)

How did the invention of the transistor transform the way communication occurs in Australia? In your answer, refer to the technology that the transistor replaced.

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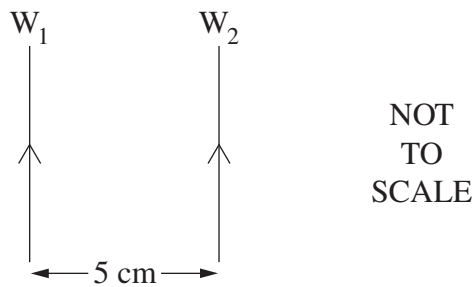
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Question 23 (6 marks)

Two identical wires, W_1 and W_2 , each 2.5 m in length, are positioned as shown. They carry identical currents in the direction indicated.



- (a) Identify the direction of the force which W_2 experiences as a result of the current in W_1 . **1**

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- (b) Calculate the current in each wire, given that the two wires experience a force of 6.9×10^{-4} N. **2**

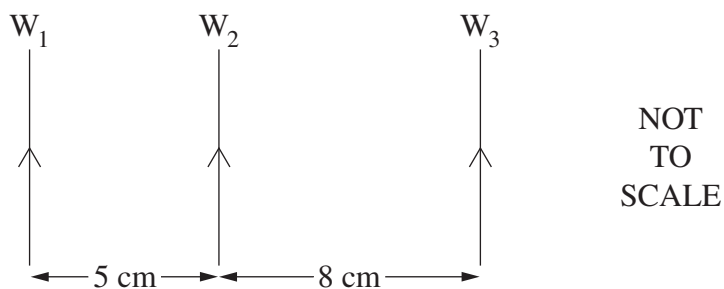
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- (c) A third wire, W_3 , carrying a smaller current, is now placed as shown. **3**



Explain qualitatively the forces on W_2 as a result of the currents in W_1 and W_3 .

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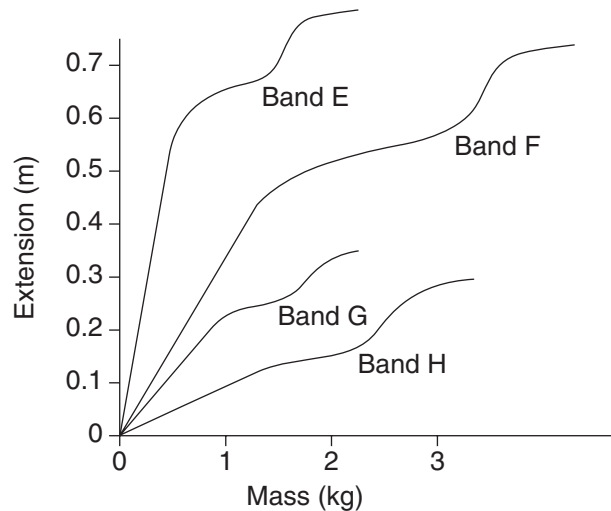
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Question 24 (4 marks)

An experiment was conducted to investigate the flexibility* and strength** of different types of rubber bands, all with the same initial length. A mass was attached to each band and the extension was measured. Masses were gradually increased, and the extensions measured until each band broke. The photograph was taken during the experiment. The results are summarised in the graph.



* Flexibility: *The more flexible the rubber band, the greater its extension for a given mass.*

** Strength: *The stronger the rubber band, the more mass it is able to hold before breaking*

- (a) Which rubber band is the most flexible? Justify your answer with reference to the graph. 2

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- (b) Identify the strongest rubber band and state the mass range in which the extension appears to be directly proportional to the attached mass. 2

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Centre Number

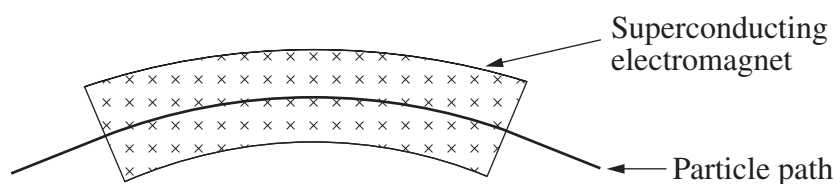
Section I – Part B (continued)

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Student Number

Question 25 (5 marks)

In the Large Hadron Collider (LHC), the particle beams are steered using magnetic fields, as shown.



- (a) Two particles with the same mass and speed are travelling through the LHC in opposite directions. **2**

What can be deduced about the charge on the particles?

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- (b) During a test run, a proton travels with a speed of $1.0 \times 10^7 \text{ m s}^{-1}$ around the LHC. The radius of curvature of its path is 4.2 m. **3**

Calculate the magnetic field strength.

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Question 26 (6 marks)

In the distribution of electricity, the overall energy losses between the power plant and users can easily be between 8% and 15%, which suggests that there is still some room to improve efficiency.

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Analyse this statement. In your analysis, you must refer to existing sources of energy loss, and a possible new technology to minimise such loss.

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Centre Number

Section I – Part B (continued)

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Student Number

Question 27 (7 marks)

In an experiment to investigate the photoelectric effect, light is shone onto a silver surface and the resulting maximum electron kinetic energy is measured and recorded.

<i>Light wavelength (nm)</i>	<i>Electron kinetic energy (eV)</i>
250	0.25
215	1.08
187	1.90
167	2.73
150	3.56

- (a) Determine the frequency of the highest energy photons used in the experiment. **2**

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- (b) What effect would changing the intensity of the light have on the measured electron kinetic energy? **1**

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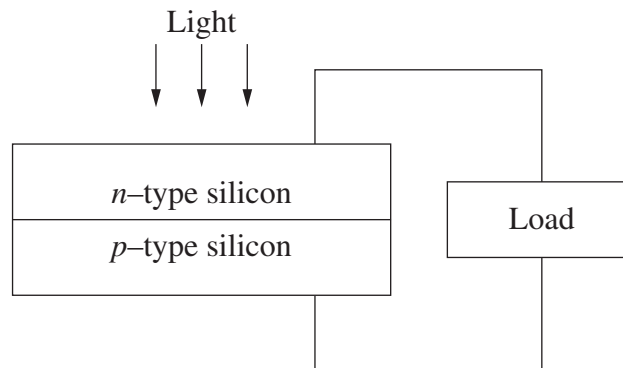
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Question 27 continues on page 22

Question 27 (continued)

- (c) With reference to the photoelectric effect, and the semiconductors shown in the diagram, explain the operation of a solar cell.

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End of Question 27

Physics

Section II

25 marks

Attempt ONE question from Questions 28–32

Allow about 45 minutes for this section

Answer the question in a writing booklet. Extra writing booklets are available.

Show all relevant working in questions involving calculations.

	Pages
Question 28 Geophysics	24–25
Question 29 Medical Physics	26–27
Question 30 Astrophysics	28–29
Question 31 From Quanta to Quarks	30–31
Question 32 The Age of Silicon	32–33

Question 28 — Geophysics (25 marks)

- (a) Radiation is used to obtain information about a surface from a distance.
- (i) Use a labelled diagram and text to outline a first-hand investigation to demonstrate the relationship between the nature of a surface and the radiation reflected. **2**
 - (ii) How would the results demonstrate the relationship between the nature of the surface and the radiation reflected? **2**
 - (iii) Use a specific example to show how radiation can be used to provide information about Earth from a distance. **2**

- (b) A pendulum was used to determine the acceleration due to gravity, g , at the base of a mountain. The period, T , was 2.00 s, and the length of the string, l , was 1.00 m.

- (i) Calculate g , using the formula, **2**

$$T = 2\pi\sqrt{\frac{l}{g}}.$$

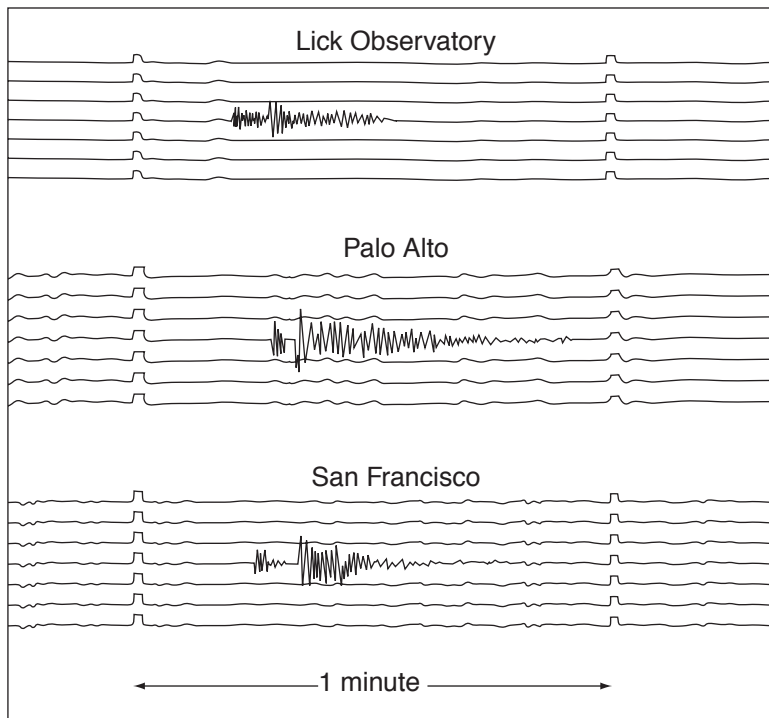
- (ii) Calculate the radius of Earth using this value of g . **2**
- (iii) The pendulum was moved to a different location where there are no mountains. The same value of T was obtained. **3**

Explain this observation.

Question 28 continues on page 25

Question 28 (continued)

- (c) Seismograms from three different geophysical stations showing the records of a small earthquake somewhere in California are shown. Time is marked on each seismogram by offsets in the records. The interval between the offsets is 1 minute.



© Department of Earth and Environmental Sciences

- (i) Explain the features of the graphs and what information can be deduced. **3**
- (ii) Describe how seismic methods are used in the search for oil and gas. **3**
- (d) Name THREE geophysical techniques, and explain how each of these provides evidence to support the claim that Earth is dynamic. **6**

End of Question 28

Question 29 — Medical Physics (25 marks)

- (a) (i) In X-ray images, the small intestine is not normally visible. **2**
- Explain how the use of a contrast medium, for example a barium meal, allows the small intestine to be seen.
- (ii) Using text and a labelled diagram, explain how X-rays are produced for medical imaging. **4**
- (b) (i) Given the velocity of sound in blood is $1.53 \times 10^3 \text{ m s}^{-1}$, and blood has a density of $1.05 \times 10^3 \text{ kg m}^{-3}$, calculate the acoustic impedance of blood. **1**
- (ii) Ultrasound can pass from blood into a variety of materials. What happens to the incident pulse when it passes into materials of increasing acoustic impedance? **1**
- (iii) Explain how a piezoelectric crystal can be used as a producer and receiver of ultrasound waves. **4**

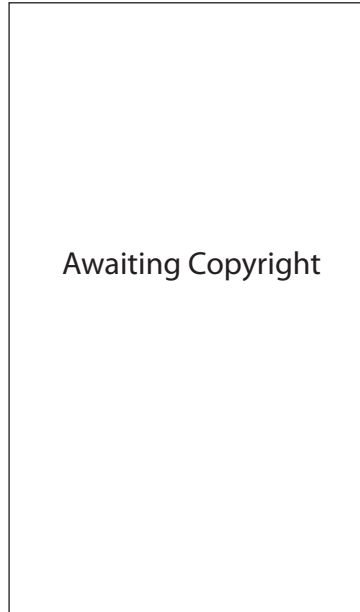
Question 29 continues on page 27

Question 29 (continued)

- (c) (i) “Have a CAT scan – live longer!” **4**

Can the claim made in this statement be justified? In your answer refer to the properties and uses of CAT scans.

- (ii) In this PET image a chemical tracer has been used to measure glucose metabolism in a patient. **3**



Explain how this image has been produced, including the physics involved.

- (d) Describe the sequence of events and associated processes of physics by which an image is produced using magnetic resonance imaging. **6**

End of Question 29

Question 30 — Astrophysics (25 marks)

- (a) (i) Distinguish between the terms *resolution* and *sensitivity* as used in astrophysics. **2**

- (ii) Interferometry and active optics are techniques that can be used to improve the resolution and/or sensitivity of ground based telescopes. **4**

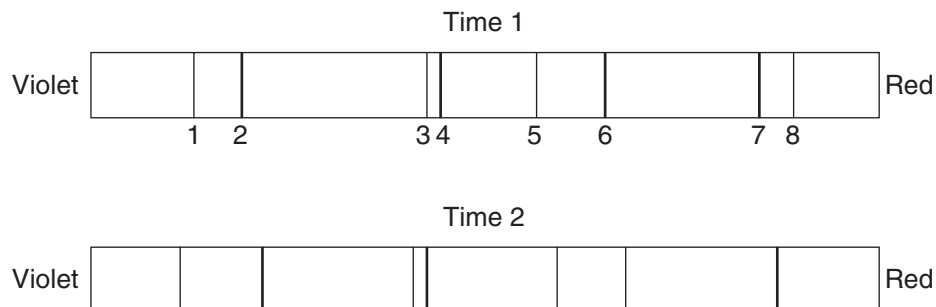
Explain why only one of these techniques is useful in improving the resolution and sensitivity of radio telescopes.

- (b) (i) Describe the modelling process used in a computer simulation which draws a light curve for an eclipsing binary star system. **2**

- (ii) Two stars in a visual binary system have an orbital period of 2.1×10^8 s and are determined to be 7.2×10^8 km apart. **2**

Calculate the combined mass of the stars.

- (iii) The spectra below show absorption lines for a variable pair of spectroscopic binary stars at two different times, Time 1 and Time 2. Each spectrum contains the absorption lines from both stars. **3**

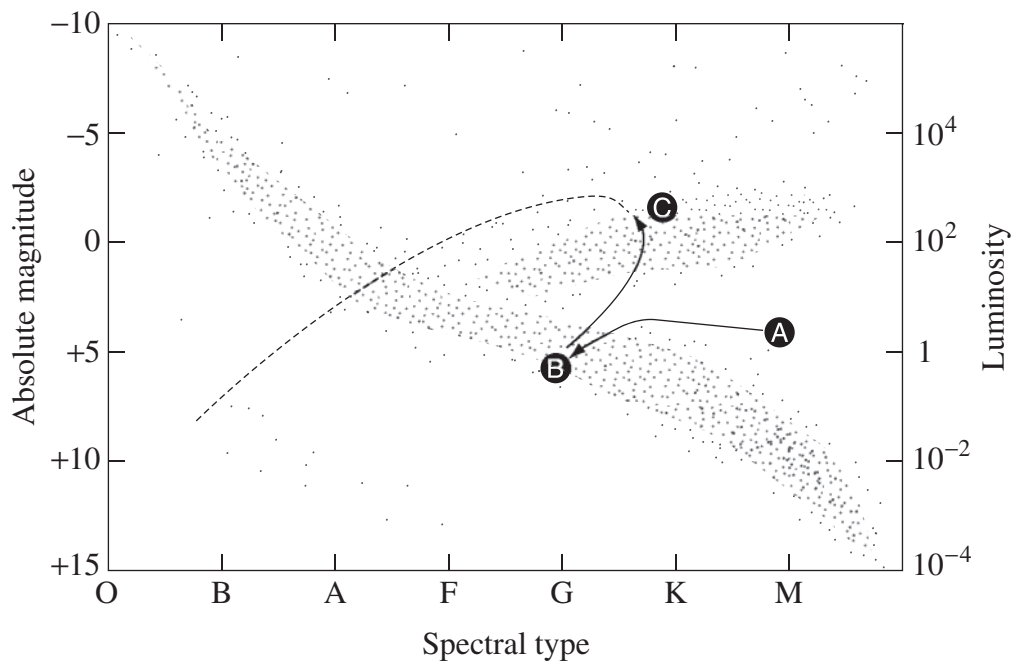


Explain why there are differences in the spectra.

Question 30 continues on page 29

Question 30 (continued)

- (c) (i) A star was found to have a visual magnitude (V) of 2.9 and a photographic magnitude (B) of 4.6. 1
- Will the star be more blue or more red in colour?
- (ii) How can the colour index (B–V) of a star be measured in an observatory? 2
- (d) Describe the advantages of using photoelectric technologies over photographic methods in photometry. 3
- (e) A possible evolutionary path of a star is shown on the Hertsprung-Russell (H-R) diagram. 6



Describe the sequence of events and the associated physical processes a star undergoes in moving from **A** to **B** to **C**.

End of Question 30

Question 31 — From Quanta to Quarks (25 marks)

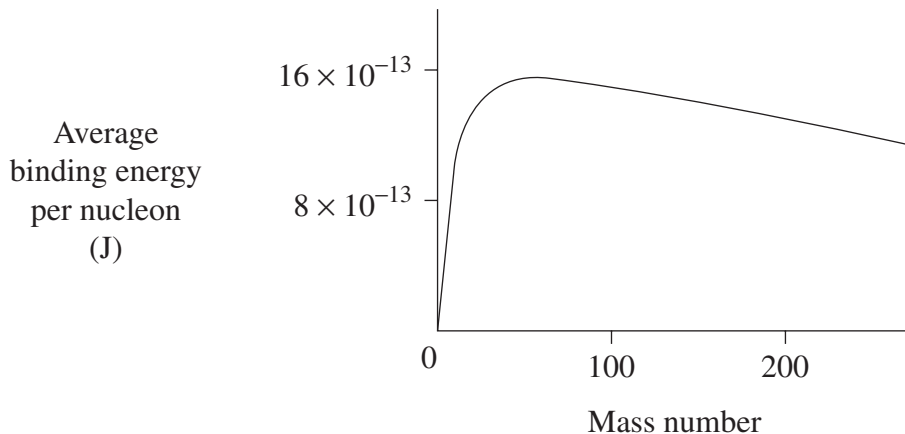
- (a) Marsden and Geiger conducted an experiment in which they fired alpha particles at a thin gold foil. Most of the particles passed straight through.
- (i) Describe how Rutherford's model of the atom explained these results. **2**
 - (ii) Describe TWO problems associated with Rutherford's model and how these were explained by Bohr's model of the hydrogen atom. **4**
- (b)
- (i) Describe de Broglie's proposal that a particle can exhibit both wave and particle properties. **2**
 - (ii) Explain how Davisson and Germer were able to confirm de Broglie's proposal. **3**
 - (iii) Calculate the velocity of an electron that has a wavelength of 3.33×10^{-10} m. **2**

Question 31 continues on page 31

Question 31 (continued)

- (c) (i) Define *mass defect*. **1**
- (ii) The energy required to separate all the nucleons within a nucleus is the binding energy. The average binding energy per nucleon is a measure of the stability of a nucleus. **2**

The graph shows how average binding energy per nucleon varies with mass number.



Use the graph to compare the stability of a nucleus of mass number 200 with a nucleus of mass number 50.

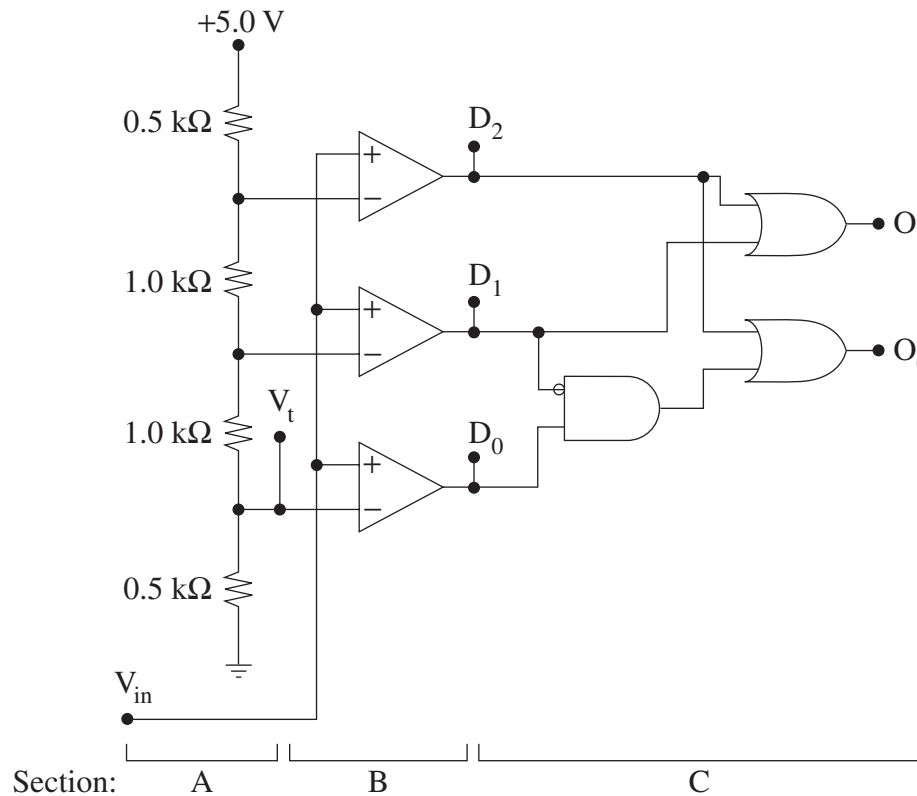
- (d) In 1920, Rutherford suggested the existence of an undiscovered nuclear particle. Explain how Chadwick confirmed Rutherford's prediction using conservation laws. **3**
- (e) Theories and experiments not only help increase our understanding but also generate new questions. **6**

Use the standard model of matter to support this statement.

End of Question 31

Question 32 — The Age of Silicon (25 marks)

- (a) The following circuit diagram shows the internal design of a 2-bit analogue to digital converter.



- | | |
|---|----------|
| (i) Identify the function of section A of the circuit. | 1 |
| (ii) Calculate the voltage V_t . | 1 |
| (iii) Are the amplifiers in section B used in an open loop or closed loop configuration? Justify your answer. | 2 |
| (iv) Construct a truth table showing the outputs O_1 and O_0 for each of the possible input states of D_0 , D_1 and D_2 . | 2 |

Question 32 continues on page 33

Question 32 (continued)

- (b) (i) Compare the function of input and output transducers. 2
- (ii) An electronic system monitors conditions in a glasshouse. The system measures sunlight intensity, temperature, and controls an electric watering pump. 5

Copy this table into your writing booklet and complete the table for this system.

	<i>Sunlight</i>	<i>Temperature</i>	<i>Pump</i>
Type of transducer			
Input or output transducer			
Outline of operation	TABLE TO BE COMPLETED IN YOUR WRITING BOOKLET		

- (c) Outline the differences between digital and analogue signals using the operation of a music CD player as an example. 2
- (d) Analyse how the development of the integrated circuit has affected energy consumption. 4
- (e) The advancement of silicon-based integrated circuit technology has resulted in progressive miniaturisation and a doubling of circuit performance roughly every 18 months. It is expected that physics limitations will soon halt this progress, requiring a reconceptualisation of the way integrated circuits are constructed. 6

If this reconceptualisation cannot be realised, what would be the likely impact on society?

End of paper

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DATA SHEET

Charge on electron, q_e	$-1.602 \times 10^{-19} \text{ C}$
Mass of electron, m_e	$9.109 \times 10^{-31} \text{ kg}$
Mass of neutron, m_n	$1.675 \times 10^{-27} \text{ kg}$
Mass of proton, m_p	$1.673 \times 10^{-27} \text{ kg}$
Speed of sound in air	340 m s^{-1}
Earth's gravitational acceleration, g	9.8 m s^{-2}
Speed of light, c	$3.00 \times 10^8 \text{ m s}^{-1}$
Magnetic force constant, $\left(k \equiv \frac{\mu_0}{2\pi} \right)$	$2.0 \times 10^{-7} \text{ N A}^{-2}$
Universal gravitational constant, G	$6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
Mass of Earth	$6.0 \times 10^{24} \text{ kg}$
Planck constant, h	$6.626 \times 10^{-34} \text{ J s}$
Rydberg constant, R (hydrogen)	$1.097 \times 10^7 \text{ m}^{-1}$
Atomic mass unit, u	$1.661 \times 10^{-27} \text{ kg}$ $931.5 \text{ MeV}/c^2$
1 eV	$1.602 \times 10^{-19} \text{ J}$
Density of water, ρ	$1.00 \times 10^3 \text{ kg m}^{-3}$
Specific heat capacity of water	$4.18 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$

FORMULAE SHEET

$$v = f\lambda$$

$$I \propto \frac{1}{d^2}$$

$$\frac{v_1}{v_2} = \frac{\sin i}{\sin r}$$

$$E = \frac{F}{q}$$

$$R = \frac{V}{I}$$

$$P = VI$$

$$\text{Energy} = VIt$$

$$v_{\text{av}} = \frac{\Delta r}{\Delta t}$$

$$a_{\text{av}} = \frac{\Delta v}{\Delta t} \text{ therefore } a_{\text{av}} = \frac{v - u}{t}$$

$$\Sigma F = ma$$

$$F = \frac{mv^2}{r}$$

$$E_k = \frac{1}{2}mv^2$$

$$W = Fs$$

$$p = mv$$

$$\text{Impulse} = Ft$$

$$E_p = -G \frac{m_1 m_2}{r}$$

$$F = mg$$

$$v_x^2 = u_x^2$$

$$v = u + at$$

$$v_y^2 = u_y^2 + 2a_y \Delta y$$

$$\Delta x = u_x t$$

$$\Delta y = u_y t + \frac{1}{2} a_y t^2$$

$$\frac{r^3}{T^2} = \frac{GM}{4\pi^2}$$

$$F = \frac{Gm_1 m_2}{d^2}$$

$$E = mc^2$$

$$l_v = l_0 \sqrt{1 - \frac{v^2}{c^2}}$$

$$t_v = \frac{t_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$m_v = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

FORMULAE SHEET

$$\frac{F}{l} = k \frac{I_1 I_2}{d}$$

$$d = \frac{1}{p}$$

$$F = BIl \sin \theta$$

$$M = m - 5 \log \left(\frac{d}{10} \right)$$

$$\tau = Fd$$

$$\frac{I_A}{I_B} = 100^{(m_B - m_A)/5}$$

$$\tau = nBIA \cos \theta$$

$$m_1 + m_2 = \frac{4\pi^2 r^3}{GT^2}$$

$$\frac{V_p}{V_s} = \frac{n_p}{n_s}$$

$$F = qvB \sin \theta$$

$$\frac{1}{\lambda} = R \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$$

$$E = \frac{V}{d}$$

$$\lambda = \frac{h}{mv}$$

$$E = hf$$

$$c = f\lambda$$

$$A_0 = \frac{V_{\text{out}}}{V_{\text{in}}}$$

$$Z = \rho v$$

$$\frac{V_{\text{out}}}{V_{\text{in}}} = - \frac{R_{\text{f}}}{R_{\text{i}}}$$

$$\frac{I_r}{I_0} = \frac{[Z_2 - Z_1]^2}{[Z_2 + Z_1]^2}$$

PERIODIC TABLE OF THE ELEMENTS

1 H 1.008 Hydrogen		KEY										2 He 4.003 Helium																							
3 Li 6.941 Lithium		4 Be 9.012 Beryllium		Atomic Number		79 Au 197.0 Gold		Symbol of element		5 B 10.81 Boron		6 C 12.01 Carbon		7 N 14.01 Nitrogen		8 O 16.00 Oxygen		9 F 19.00 Fluorine		10 Ne 20.18 Neon															
		12 Mg 24.31 Magnesium		Atomic Weight		Name of element				13 Al 26.98 Aluminium		14 Si 28.09 Silicon		15 P 30.97 Phosphorus		16 S 32.07 Sulfur		17 Cl 35.45 Chlorine		18 Ar 39.95 Argon															
19 K 39.10 Potassium		20 Ca 40.08 Calcium		21 Sc 44.96 Scandium		22 Ti 47.87 Titanium		23 V 50.94 Vanadium		24 Cr 52.00 Chromium		25 Mn 54.94 Manganese		26 Fe 55.85 Iron		27 Co 58.93 Cobalt		28 Ni 58.69 Nickel		29 Cu 63.55 Copper		30 Zn 65.41 Zinc		31 Ga 69.72 Gallium		32 Ge 72.64 Germanium		33 As 74.92 Arsenic		34 Se 78.96 Selenium		35 Br 79.90 Bromine		36 Kr 83.80 Krypton	
37 Rb 85.47 Rubidium		38 Sr 87.62 Strontium		39 Y 88.91 Yttrium		40 Zr 91.22 Zirconium		41 Nb 92.91 Niobium		42 Mo 95.94 Molybdenum		43 Tc [97.91] Technetium		44 Ru 101.1 Ruthenium		45 Rh 102.9 Rhodium		46 Pd 106.4 Palladium		47 Ag 107.9 Silver		48 Cd 112.4 Cadmium		49 In 114.8 Indium		50 Sn 118.7 Tin		51 Sb 121.8 Antimony		52 Te 127.6 Tellurium		53 I 126.9 Iodine		54 Xe 131.3 Xenon	
55 Cs 132.9 Caesium		56 Ba 137.3 Barium		57–71 Lanthanoids		72 Hf 178.5 Hafnium		73 Ta 180.9 Tantalum		74 W 183.8 Tungsten		75 Re 186.2 Rhenium		76 Os 190.2 Osmium		77 Ir 192.2 Iridium		78 Pt 195.1 Platinum		79 Au 197.0 Gold		80 Hg 200.6 Mercury		81 Tl 204.4 Thallium		82 Pb 207.2 Lead		83 Bi 209.0 Bismuth		84 Po [209.0] Polonium		85 At [210.0] Astatine		86 Rn [222.0] Radon	
87 Fr [223] Francium		88 Ra [226] Radium		89–103 Actinoids		104 Rf [261] Rutherfordium		105 Db [262] Dubnium		106 Sg [266] Seaborgium		107 Bh [264] Bohrium		108 Hs [277] Hassium		109 Mt [268] Meitnerium		110 Ds [271] Darmstadtium		111 Rg [272] Roentgenium															

Lanthanoids

57 La 138.9 Lanthanum	58 Ce 140.1 Cerium	59 Pr 140.9 Praseodymium	60 Nd 144.2 Neodymium	61 Pm [145] Promethium	62 Sm 150.4 Samarium	63 Eu 152.0 Europium	64 Gd 157.3 Gadolinium	65 Tb 158.9 Terbium	66 Dy 162.5 Dysprosium	67 Ho 164.9 Holmium	68 Er 167.3 Erbium	69 Tm 168.9 Thulium	70 Yb 173.0 Ytterbium	71 Lu 175.0 Lutetium
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Actinoids

89 Ac [227] Actinium	90 Th 232.0 Thorium	91 Pa 231.0 Protactinium	92 U 238.0 Uranium	93 Np [237] Neptunium	94 Pu [244] Plutonium	95 Am [243] Americium	96 Cm [247] Curium	97 Bk [247] Berkelium	98 Cf [251] Californium	99 Es [252] Einsteinium	100 Fm [257] Fermium	101 Md [258] Mendelevium	102 No [259] Nobelium	103 Lr [262] Lawrencium
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For elements that have no stable or long-lived nuclides, the mass number of the nuclide with the longest confirmed half-life is listed between square brackets.
The International Union of Pure and Applied Chemistry Periodic Table of the Elements (October 2005 version) is the principal source of data. Some data may have been modified.



2009 HSC Physics Marking Guidelines

Section I, Part A

Question	Response
1	C
2	A
3	C
4	D
5	C
6	A
7	C
8	C
9	A
10	B
11	B
12	D
13	B
14	C
15	A

Section I, Part B**Question 16 (a)***Outcomes assessed: H6, H9, H12***MARKING GUIDELINES**

Criteria	Marks
• Correctly calculates weight of space craft on Earth	1

Question 16 (b)*Outcomes assessed: H6, H9, H12***MARKING GUIDELINES**

Criteria	Marks
• Substitutes correctly to calculate mass	2
• Uses formula correctly, but makes an error in substitution	1

Question 17 (a)

Outcomes assessed: H6, H11, H13, H14

MARKING GUIDELINES

Criteria	Marks
- Draws diagrams of a valid experimental set up - Uses labels on the diagrams to show how the experimental set up enabled a distinction to be made between inertial and non-inertial frames of reference OR - Draws a labelled diagram and describes the difference between inertial and non-inertial frames of reference	2
- Draws a diagram of a valid experimental set up OR - States how to distinguish between an inertial and non-inertial frame of reference	1

Question 17 (b)

Outcomes assessed: H6, H14

MARKING GUIDELINES

Criteria	Marks
- Describes the major principles of relativity - Relates how the principles of relativity applies for inertial frames of reference	3
- Describes some features of relativity OR - Defines inertial and non-inertial frames of reference OR - Defines an inertial OR non-inertial frame of reference and links to a principle of relativity	2
Makes a general statement about EITHER the theory of special relativity OR inertial frames of reference	1

Question 18 (a)

Outcomes assessed: H12, H13, H14

MARKING GUIDELINES

Criteria	Marks
- Substitutes correctly to calculate distance in light years OR - Substitutes correctly to calculate distance in SI units	2
Uses formula correctly but makes an error in substitution	1

Question 18 (b)

Outcomes assessed: H12, H13, H14

MARKING GUIDELINES

Criteria	Marks
- Substitutes correctly to calculate time in light years OR - Substitutes correctly to calculate time in SI units	2
Uses formula correctly but makes an error in substitution	1

Question 19 (a)

Outcomes assessed: H6, H9, H12, H13

MARKING GUIDELINES

Criteria	Marks
Substitutes correctly to determine the force	3
- Combines TWO appropriate equations AND EITHER - Makes an error in rearranging the equations OR - Makes an error in substitution	2
Chooses one correct formula and attempts to find the force or electric field	1

Question 19 (b)

Outcomes assessed: H6, H12, H13, H14

MARKING GUIDELINES

Criteria	Marks
Substitutes correctly to determine time	3
- Attempts any TWO of the following - Substitutes correctly to determine acceleration - Substitutes correctly to determine horizontal component of velocity - Chooses the correct formula and attempts to find time	2
- Chooses ONE correct formula only and attempts to find the acceleration OR - Chooses ONE correct formula only and attempts to find the horizontal component of velocity OR - Chooses ONE correct formula only and attempts to find time	1

Question 20

Outcomes assessed: H7, H13

MARKING GUIDELINES

Criteria	Marks
- Identifies THREE appliances - Identifies correctly the THREE different energy transfers/transformations - Identifies a use for each appliance - Uses an appropriate table to display information	4
- Provides THREE of the following: - Identifies THREE appliances - Identifies correctly the THREE different energy transfers/transformations - Identifies a use for each appliance - Uses an appropriate table to display information OR - Completes the table for two appliances	3
- Provides TWO of the following: - Identifies THREE appliances - Identifies THREE energy transfers/transformations with different energy outputs correctly - Identifies a use for each appliance - Uses an appropriate table to display information OR - Completes the table for one appliance	2
- Draws an appropriate table and correctly fills in some cells OR - Identifies an energy transformation for an appliance OR - Outlines the features of THREE appliances	1

Question 21 (a)

Outcomes assessed: H9, H14

MARKING GUIDELINES

Criteria	Marks
Identifies correct side of loop	1

Question 21 (b)

Outcomes assessed: H9, H12, H13, H14

MARKING GUIDELINES

Criteria	Marks
Substitutes correctly to calculate torque	2
Uses formula correctly, but makes an error in substitution	1

Question 21 (c)

Outcomes assessed: H9, H12, H13, H14

MARKING GUIDELINES

Criteria	Marks
Manipulates equations to calculate the value of magnetic field strength	3
Equates two appropriate equations, AND EITHER - Makes an error in rearranging the equations OR - Makes an error in substitution	2
- Identifies an appropriate equation - Attempts to calculate magnetic field strength	1

Question 22

Outcomes assessed: H4, H13, H14

MARKING GUIDELINES

Criteria	Marks
• Demonstrates a sound knowledge of the development of the transistor as a replacement for thermionic devices • Describes the advantages of using solid state devices over thermionic devices • Identifies ways in which communication has been transformed in Australia • Demonstrates coherence and logical progression within the answer	4
• Identifies ways in which communication has been transformed in Australia • Demonstrates some knowledge about the development of solid state devices OR • States some of the advantages of using solid state devices over thermionic devices	2–3
• Lists some advantages of solid state devices over thermionic devices OR • States what a transistor does in a circuit OR • Lists some communication devices which use transistors or integrated circuits	1

Question 23 (a)

Outcomes assessed: H9, H14

MARKING GUIDELINES

Criteria	Marks
Identifies correct direction	1

Question 23 (b)

Outcomes assessed: H9, H12, H13, H14

MARKING GUIDELINES

Criteria	Marks
Substitutes correctly to calculate current	2
Identifies the formula correctly but makes an error in substitution	1

Question 23 (c)

Outcomes assessed: H9, H14

MARKING GUIDELINES

Criteria	Marks
- Identifies the force on W_2 due to W_1 and W_3 relating the magnitude and direction of each force to distance and current - Describes the resultant force on W_2	3
Identifies the force on W_2 due to W_1 and W_3 relating the magnitude of each force to distance and current	2
Describes the force on W_2 due to W_1 and W_3	1

Question 24 (a)

Outcomes assessed: H9, H12, H14

MARKING GUIDELINES

Criteria	Marks
- Correctly identifies the most flexible rubber band - Provides a valid reason why that rubber band is most flexible	2
Correctly identifies the most flexible rubber band	1

Question 24 (b)

Outcomes assessed: H12, H14

MARKING GUIDELINES

Criteria	Marks
- Correctly identifies the strongest band - Correctly identifies the region in which extension is directly proportional to the attached mass	2
- Correctly identifies the strongest band OR - Correctly identifies the region in which extension is directly proportional to the attached mass	1

Question 25 (a)

Outcomes assessed: H9, H10, H14

MARKING GUIDELINES

Criteria	Marks
Correctly identifies the type of charge and qualitatively states its magnitude	2
- Correctly identifies the type of charge OR - Qualitatively states the magnitude of the charge	1

Question 25 (b)

Outcomes assessed: H9, H10, H12, H13, H14

MARKING GUIDELINES

Criteria	Marks
Substitutes correctly to determine the magnetic field strength	3
- Equates two appropriate equations - Makes an error in rearranging the equations	2
Chooses one correct formula only and attempts to find the magnetic field strength	1

Question 26

Outcomes assessed: H7, H9, H14

MARKING GUIDELINES

Criteria	Marks
• Demonstrates a thorough knowledge of how energy losses occur in transmission • Describes present techniques to reduce energy losses and relates to energy losses in transmission • Identifies a new technology to reduce energy loss and relates to how this technology could minimise energy losses in a source • Demonstrates coherence and logical progression	6
• Demonstrates a sound knowledge of how energy losses occur in transmission • Describes present techniques to reduce energy losses • Identifies a new technology to reduce energy loss and provides a feature of this technology • Communicates some scientific principles and ideas in a clear manner	4–5
• Demonstrates a knowledge of how energy is lost in transmission and identifies a new technology to reduce the losses • Communicates ideas in a basic form using general scientific terms OR • Demonstrates a sound knowledge about how energy is lost in transmission and states how such loss is minimised • Communicates ideas in a basic form using general scientific terms	2–3
• Communicates simple ideas AND EITHER • States how energy is lost through a source OR • Identifies a new technology to reduce energy losses OR • Identifies how an energy loss is presently minimised OR • Identifies two sources of energy loss	1

Question 27 (a)

Outcomes assessed: H7, H10, H12, H13

MARKING GUIDELINES

Criteria	Marks
• Substitutes correctly to calculate the frequency	2
• Uses formula correctly but makes an error in substitution OR • Makes an error in rearranging the equation	1

Question 27 (b)

Outcomes assessed: H7, H10, H14

MARKING GUIDELINES

Criteria	Marks
• Correctly identifies the effect	1

Question 27 (c)

Outcomes assessed: H7, H10

MARKING GUIDELINES

Criteria	Marks
• Describes the photoelectric effect • Relates the description of the photoelectric effect and movement of charge at a p - n junction to the operation of a solar cell • Describes the flow of charge in p and n type semiconductors	4
• Describes the photoelectric effect AND/OR • Describes the flow of charge in p and n type semiconductors	2–3
• Outlines the photoelectric effect OR • Outlines the structure of p and n type semiconductors	1

Section II

Question 28 (a) (i)

Outcomes assessed: H10, H11

MARKING GUIDELINES

Criteria	Marks
- Uses labels and text to demonstrate relevant features - Draws a diagram of a valid experimental set up	2
Draws a diagram of a valid experimental set up	1

Question 28 (a) (ii)

Outcomes assessed: H10

MARKING GUIDELINES

Criteria	Marks
- Uses results to demonstrate relationship - Outlines results	2
States results	1

Question 28 (a) (iii)

Outcomes assessed: H10

MARKING GUIDELINES

Criteria	Marks
Outlines how radiation can be used for remote sensing using one specific example	2
States an example of using radiation for remote sensing	1

Question 28 (b) (i)

Outcomes assessed: H9, H12, H14

MARKING GUIDELINES

Criteria	Marks
• Substitutes correctly to calculate g	2
• Uses formula correctly but makes an error in substitution	1

Question 28 (b) (ii)

Outcomes assessed: H9, H12, H14

MARKING GUIDELINES

Criteria	Marks
• Substitutes correctly to calculate radius of Earth	2
• Uses formula correctly but makes an error in substitution	1

Question 28 (b) (iii)

Outcomes assessed: H9, H14

MARKING GUIDELINES

Criteria	Marks
• Relates to geophysical situation • Relate the features affecting period to the measured value • Identifies features affecting the period	3
• Outlines features affecting the period • Outlines geophysical situation	2
• States a relevant and correct geophysical feature OR • States one feature affecting period	1

Question 28 (c) (i)

Outcomes assessed: H13, H14

MARKING GUIDELINES

Criteria	Marks
• Outlines features on graph relating to S and P waves • Relates features on graph to delay between S and P waves • Outlines how the delay can be used to deduce the location of the epicentre	3
• Outlines features about S and P waves • Indicates how to deduce location of epicentre	2
• States correct information about S and P waves with reference to graph	1

Question 28 (c) (ii)

Outcomes assessed: H13, H14

MARKING GUIDELINES

Criteria	Marks
• Provides features of seismic methods • Relates features to search for oil and gas	2–3
• Outlines seismic methods	1

Question 28 (d)

Outcomes assessed: H13, H14

MARKING GUIDELINES

Criteria	Marks
• Demonstrates a thorough knowledge of geophysical techniques and dynamic nature of Earth • Describes THREE geophysical techniques • Relates each geophysical technique to theories supporting the changing nature of Earth • Use evidence and examples as appropriate • Demonstrates coherence and logical progression and includes correct use of scientific principles and ideas	6
• Demonstrates a sound knowledge of geophysical techniques • Outlines TWO geophysical techniques and relates each to theories supporting the changing nature of Earth • Uses some evidence or examples • Communicates some scientific principles and ideas in a clear manner	4–5
• Demonstrates a basic knowledge of geophysical techniques • Communicates ideas in a basic form • Outlines a geophysical technique and attempts to relate to the changing nature of Earth	2–3
• Outlines changing nature of Earth • Communicates simple ideas OR • Identifies a geophysical technique • Communicates simple ideas	1

Question 29 (a) (i)

Outcomes assessed: H10

MARKING GUIDELINES

Criteria	Marks
- Identifies the need for a material with higher absorption - Relates the higher absorption of x-rays to contrast	2
- Identifies the need for a material with higher absorption OR - Relates the higher absorption of x-rays to contrast	1

Question 29 (a) (ii)

Outcomes assessed: H10, H13

MARKING GUIDELINES

Criteria	Marks
- Provides a valid and clearly labelled diagram - Uses explicit text to describe what happens in the tube and the atomic processes	4
- Provides a simple but valid diagram with some labels and text referring to what happens in the tube and/or atomic physics OR - Uses explicit text to describe what happens in the tube and the atomic processes	2-3
- Provides a labelled diagram of an x-ray tube OR - Uses some text to describe either the processes in the tube or the atomic physics	1

Question 29 (b) (i)

Outcomes assessed: H8, H12, H13

MARKING GUIDELINES

Criteria	Marks
Correctly calculates the acoustic impedance of blood	1

Question 29 (b) (ii)

Outcomes assessed: H8

MARKING GUIDELINES

Criteria	Marks
Correctly states the effect on the incident pulse	1

Question 29 (b) (iii)

Outcomes assessed: H8

MARKING GUIDELINES

Criteria	Marks
- Provides a description of a piezoelectric crystal - Relates changes in the crystal to the production of ultrasound waves - Relates ultrasound waves to the production of an electric potential	4
- Outlines a piezoelectric crystal - Relates changing electric potential to the production of ultrasound - Relates ultrasound waves to distortion of the piezoelectric crystal	2-3
- Outlines a piezoelectric crystal OR - Relates changing electric potential to the production of ultrasound waves OR - Relates ultrasound waves to the distortion of the piezoelectric crystal	1

Question 29 (c) (i)

Outcomes assessed: H10

MARKING GUIDELINES

Criteria	Marks
Provides detailed arguments for or against the use of CAT scans including explicit examples of its use	4
Outlines some points for or against the use/properties of CAT scans	2–3
- Identifies ONE advantage of using a CAT scan OR - Identifies ONE disadvantage of using a CAT scan OR - States ONE use/property of a CAT scan	1

Question 29 (c) (ii)

Outcomes assessed: H10, H12

MARKING GUIDELINES

Criteria	Marks
- Relates the use of a tracer to accumulation in the target organs - Provides a description of electron-positron annihilation to produce gamma rays - Provides a description of how gamma rays are detected	3
- Provides TWO of the following: - Outlines the role of the tracer - Provides an outline of positron emission and the production of gamma rays - Provides an outline of the detection of the gamma rays	2
Provides ONE piece of relevant information	1

Question 29 (d)

Outcomes assessed: H8, H9, H13, H14

MARKING GUIDELINES

Criteria	Marks
• Demonstrates thorough knowledge and understanding of producing an MRI image • Provides a detailed sequence of events involved with specific reference to the physics involved • Demonstrates coherence and logical progression and includes correct use of scientific terms	5–6
• Demonstrates sound knowledge and understanding of producing an MRI image • Communicates some scientific principles and ideas in a clear manner AND EITHER • Provides a detailed sequence of events with some reference to the physics involved OR • Provides a simple sequence of events and demonstrates sound knowledge of the physics involved in these	3–4
• Demonstrates a basic knowledge of MRI images • Communicates simple ideas AND EITHER • Provides a simple sequence of events with no reference to the physics involved OR • Provides one to two steps with some physics involved	1–2

Question 30 (a) (i)

Outcomes assessed: H10

MARKING GUIDELINES

Criteria	Marks
• Clearly notes the differences between resolution and sensitivity	2
• Identifies resolution or sensitivity	1

Question 30 (a) (ii)

Outcomes assessed: H8, H10

MARKING GUIDELINES

Criteria	Marks
- Describes both active optics and interferometry - Identifies how interferometry improves resolution and sensitivity - Identifies reason why active optics are not useful	4
- Does above but OMITs - Description of active optics OR - How interferometry improves resolution and sensitivity OR - Does not identify why active optics is not useful	3
Omits 2 of the above	2
- Only describes active optics or interferometry OR - States how interferometry improves resolution and sensitivity OR - Identifies reason why active optics is not useful	1

Question 30 (b) (i)

Outcomes assessed: H8, H10

MARKING GUIDELINES

Criteria	Marks
- Provides features of what is modelled with the computer simulation - Relates the light curve to the model	2
- Describes a light curve OR - Describes the binary star being modelled	1

Question 30 (b) (ii)

Outcomes assessed: H12, H13

MARKING GUIDELINES

Criteria	Marks
Substitutes correctly to calculate the combined mass of the stars	2
Identifies the formula correctly but makes an error in substitution	1

Question 30 (b) (iii)

Outcomes assessed: H14

MARKING GUIDELINES

Criteria	Marks
Correctly identifies TWO differences and gives reasons for the differences	3
Correctly identifies TWO differences and gives one reason for the differences	2
- Identifies TWO differences OR - Describes the reasons for the differences	1

Question 30 (c) (i)

Outcomes assessed: H13, H14

MARKING GUIDELINES

Criteria	Marks
Identifies star colour correctly	1

Question 30 (c) (ii)

Outcomes assessed: H13, H14

MARKING GUIDELINES

Criteria	Marks
Describes how B and V are measured and relates them to colour index	2
- Describes how the magnitude of a star can be measured OR - States how B OR V are measured	1

Question 30 (d)

Outcomes assessed: H2

MARKING GUIDELINES

Criteria	Marks
Describes advantages of photoelectric technologies	3
- Identifies advantages - Identifies photoelectric technologies and photographic methods OR - Describes advantages	2
Identifies an advantage	1

Question 30 (e)

Outcomes assessed: H14

MARKING GUIDELINES

Criteria	Marks
• Demonstrates a thorough knowledge and understanding of the evolutionary path of the star • Provides a detailed sequence of events • Demonstrates thorough knowledge of the physical processes • Demonstrates coherence and logical progression and includes correct use of scientific principles and ideas	5–6
• Demonstrates a sound knowledge and understanding of the evolutionary path of the star • Communicates some scientific principles and ideas in a clear manner • Provides detailed sequence of events with some reference to the physical process OR • Provides a simple sequence of events and demonstrates sound knowledge of the physical processes involved in these	3–4
• Demonstrates a basic knowledge of the evolutionary path of a star • Communicates simple ideas • Provides simple sequence of events with no reference to the physical processes involved OR • Provides one to two steps with some physical processes involved	1–2

Question 31 (a) (i)

Outcomes assessed: H2

MARKING GUIDELINES

Criteria	Marks
• Outlines TWO features of Rutherford's model that explain the experimental results	2
• Outlines ONE feature of Rutherford's model that explains the experimental results	1

Question 31 (a) (ii)

Outcomes assessed: H10

MARKING GUIDELINES

Criteria	Marks
• Describes TWO problems associated with Rutherford's model of the atom • Relates how Bohr's model of the atom was able to deal with the problems associated with Rutherford's model	4
• Describes ONE or TWO problems associated with Rutherford's model of the atom • Relates how Bohr's model of the atom was able to deal with ONE problem associated with Rutherford's model	2-3
• Describes a problem associated with Rutherford's model of the atom OR • Outline Bohr's model of the atom	1

Question 31 (b) (i)

Outcomes assessed: H8, H10

MARKING GUIDELINES

Criteria	Marks
Provides features and characteristics of de Broglie's concept of matter waves	2
Identifies ONE feature of de Broglie's proposal	1

Question 31 (b) (ii)

Outcomes assessed: H8, H10

MARKING GUIDELINES

Criteria	Marks
- Outlines the experimental procedure carried out by Davidson and Germer - Identifies that electrons showed diffraction during the experiments - Relates the display of diffraction to a wave-like property	3
- Outlines the experimental procedure carried out by Davidson and Germer - Identifies that electrons showed diffraction during the experiments OR - Relates the display of diffraction to a wave-like property	2
- Outlines the experimental procedure carried out by Davidson and Germer OR - Identifies that electrons showed diffraction during experiments OR - Relates the display of diffraction to a wave-like property	1

Question 31 (b) (iii)

Outcomes assessed: H8, H10, H12

MARKING GUIDELINES

Criteria	Marks
Substitutes correctly to calculate velocity	2
Uses formula correctly but makes an error in substitution	1

Question 31 (c) (i)

Outcomes assessed: H7, H8, H10, H12, H14

MARKING GUIDELINES

Criteria	Marks
Correctly defines the term mass defect	1

Question 31 (c) (ii)

Outcomes assessed: H8, H10, H12, H14

MARKING GUIDELINES

Criteria	Marks
- Identifies that the binding energy of the nucleus with $A=200$ is less than the binding energy of nucleus with $A=50$ - Indicates the relationship between binding energy and stability of a nucleus	2
- Identifies that the binding energy of the nucleus with $A=200$ is less than the binding energy of nucleus with $A=50$ OR - Indicates the relationship between binding energy and stability of a nucleus	1

Question 31 (d)

Outcomes assessed: H8, H10

MARKING GUIDELINES

Criteria	Marks
- Describes Chadwick's experimental observations which lead him to conclude that an additional particle/radiation was emitted - Relates how Chadwick used the laws of observation of momentum and kinetic energy to deduce the particle nature of the neutron	3
- Outlines Chadwick's experimental observations which lead him to conclude that an additional particle/radiation was emitted - Relates how Chadwick used the laws of observation of momentum and kinetic energy to deduce the particle nature of the neutron	2
- Outlines Chadwick's experiments concerning the existence of an additional particle within the nucleus OR - States one property of a neutron	1

Question 31 (e)

Outcomes assessed: H1, H8, H10, H14

MARKING GUIDELINES

Criteria	Marks
• Demonstrates thorough knowledge and understanding of the standard model of matter • Describes a theory or experiment that generated a new question or supported a postulate or prediction • Relates contribution of the standard model to our understanding of matter • Demonstrates coherence and logical progression and includes correct use of scientific principles and ideas	5–6
• Demonstrates sound knowledge and understanding of the standard model of matter • Outlines how the standard model contributes to our understanding of matter • Communicates some scientific principles and ideas in a clear manner	3–4
• Communicates ideas in a basic form using general scientific terms AND • Describe how theories and/or experiments have lead to an increased understanding of matter OR • Demonstrates a basic knowledge of the standard model of matter	1–2

Question 32 (a) (i)

Outcomes assessed: H13, H14

MARKING GUIDELINES

Criteria	Marks
Correctly identifies Section A	1

Question 32 (a) (ii)

Outcomes assessed: H12, H13

MARKING GUIDELINES

Criteria	Marks
Correctly calculates the potential V_t	1

Question 32 (a) (iii)

Outcomes assessed: H9, H14

MARKING GUIDELINES

Criteria	Marks
- Correctly identifies open loop configuration - Justifies that no feedback mechanism is present in the amplifier circuit	2
- Correctly identifies open loop configuration OR - Correctly observes that no feedback is used in the amplifier circuit	1

Question 32 (a) (iv)

Outcomes assessed: H9, H14

MARKING GUIDELINES

Criteria	Marks
- Draws correctly structured truth table - Correctly completes at least 3 terms	2
- Draws correctly structured truth table OR - Draws a badly structured truth table and gets at least 2 terms of this table correct	1

Question 32 (b) (i)

Outcomes assessed: H9, H14

MARKING GUIDELINES

Criteria	Marks
- Correctly identifies the function of a transducer - Shows similarities or differences of the input/output nature of each	2
Makes a correct statement about the function of transducers	1

Question 32 (b) (ii)

Outcomes assessed: H7

MARKING GUIDELINES

Criteria	Marks
Correctly identifies all three transducers, their type and operation	5
- Correctly identifies all three transducers - Correctly outlines the operation of at least one	3-4
- Correctly identifies at least two transducers and their type - Correctly outlines the operation of at least one	2
Correctly identifies at least one transducer and its type	1

Question 32 (c)

Outcomes assessed: H7

MARKING GUIDELINES

Criteria	Marks
- Correctly differentiates between analogue and digital signals - Correctly identifies the use of analogue and digital signals in a CD player	2
- Makes a correct statement about the characteristics of analogue or digital signals OR - Makes a correct statement about analogue or digital signals in a CD player	1

Question 32 (d)

Outcomes assessed: H4

MARKING GUIDELINES

Criteria	Marks
- Identifies features of IC relevant to energy consumption - Relates IC to energy consumption - Draws out implications	3-4
- Identifies features of IC relevant to energy consumption OR - Identifies implications	1-2

Question 32 (e)

Outcomes assessed: H3, H14

MARKING GUIDELINES

Criteria	Marks
- Demonstrates a thorough knowledge and understanding of integrated circuits - Provides a detailed description of advancements in silicon-based IC - Relates above to miniaturisation - Relates both of above to physics limitations (analogue and digital) - Discusses current society and what will, and will not change, if reconceptualisation is not realised - Demonstrates coherence and logical progression and includes correct use of scientific principles and ideas	5-6
- Demonstrates a sound knowledge and understanding of integrated circuits - Provides a description of advancements in silicon-based IC and relates to miniaturisation and physics limitations - Discusses above with some reference to current society and reconceptualisation - Communicates some scientific principles and ideas in a clear manner	3-4
- Demonstrates a basic knowledge of integrated circuits - Communicates simple ideas AND EITHER - Provides an outline at advances in IC OR - Attempts to relate in a valid manner, to reconceptualisation and current society	1-2

Physics

2009 HSC Examination Mapping Grid

Question	Marks	Content	Syllabus outcomes
Section I			
Part A			
1	1	9.2.3.2.1	H9
2	1	9.2.2.3.4, 14.1f, 12.4b	H9, H14, H12
3	1	9.2.1.2.3	H6, H7, H9
4	1	9.2.2.2.1	H6, H9, H14
5	1	9.2.3.2.3	H9
6	1	9.3.3.2.1	H9
7	1	9.3.2.2.7, 14.1d	H9, H14
8	1	9.4.1.2.5, 14.1d	H9, H14.d
9	1	9.3.4.2.2	H9
10	1	9.4.3.2.5, 9.4.3.3.2, 12e	H3, H12
11	1	9.3.1.2.3, 9.3.1.2.5, 13	H9, H13.e
12	1	9.4.3.2.7, 9.4.3.2.2, 14.1f	H9, H14
13	1	9.4.4.2.5	H9
14	1	9.4.4.2.5, 9.4.2.2.4, 9.4.2.3.2	H9
15	1	9.4.1.2.4, 9.4.1.2.6, 13.1e	H9, H13
Section I			
Part B			
16 (a)	1	9.2.1.3.3, 12.4b	H6, H9, H12
16 (b)	2	9.2.2.3.5, 9.2.3.3.2, 9.2.2.3.4, 12.4b	H6, H9, H12
17 (a)	2	9.2.4.3.2, 11.3a, 13.1a, 14.1f	H6, H11, H13, H14
17 (b)	3	9.2.4.2.4, 9.2.4.2.5, 14	H6, H14
18 (a)	2	9.2.4.3.5, 13.1d, 14.1f, 12.4b	H12, H13, H14
18 (b)	2	9.2.4.3.5, 9.2.2.3.1, 13.1d, 14.1f, 12.4b	H12, H13, H14
19 (a)	3	9.4.1.3.3, 9.2.2.3.1, 12.4b, 13.1d	H6, H9, H12, H13
19 (b)	3	9.2.2.3.2, 9.2.2.3.1, 12.4b, 13.1d, 14.1f, 14.3b	H6, H12, H13, H14
20	4	9.3.5.3.2, 9.3.1.3.5, 9.3.4.2.6, 13.1a	H7, H13
21 (a)	1	9.3.1.2.1, 14.1f	H9, H14
21 (b)	2	9.3.1.2.3, 12.4b, 13.1d, 14.1f	H9, H12, H13, H14
21 (c)	3	9.3.1.2.3, 9.3.1.3.3, 12.4b, 13.1d, 14.1f	H9, H12, H13, H14
22	4	9.4.3.3.2, 9.4.3.3.3., 13.1a, 14.3b	H4, H13, H14
23 (a)	1	9.3.1.2.2, 14.1d	H9, H14
23 (b)	2	9.3.1.3.1, 12.4b, 13.1d	H9, H12, H13, H14
23 (c)	3	9.3.1.2.2, 13.1, 14.1f, 14.1d,g	H9, H14
24 (a)	2	12.3c, 12.4c, 14.1a, 14.1d, 14.3e	H9, H12, H14
24 (b)	2	12.3c, 12.4e, 14.1a, 14.1d, 14.3e	H12, H14

Question	Marks	Content	Syllabus outcomes
25 (a)	2	9.4.1.2.3, 14.1d, 14.3c	H9, H10, H14
25 (b)	3	9.4.1.3.3, 9.2.2.3.4, 9.4.1.2.5, 12.4b, 13.1d, 14.1f	H9, H10, H12, H13, H14
26	6	9.3.5.3.2, 9.4.4.2.7, 9.4.4.3.5, 9.3.4.3.3, 14.3b	H7, H9, H14
27 (a)	2	9.4.2.2.6, 9.4.2.3.4, 12.4b	H7, H10, H12, H13
27 (b)	1	9.4.2.3.2, 9.4.2.2.4, 14.1d	H7, H10, H14
27 (c)	4	9.4.2.3.3	H7, H10

Section II

Question 28 — Geophysics

28 (a) (i)	2	9.5.2.3.1	H10, H11
28 (a) (ii)	2	9.5.2.2.1	H10
28 (a) (iii)	2	9.5.2.2.3	H10
28 (b) (i)	2	9.5.1.3.1, 12.4b, 14.1d	H9, H12, H14
28 (b) (ii)	2	9.5.2.3.3, 9.5.2.3.4, 9.2.3.3.2, 12.4b, 14.1d	H9, H12, H14
28 (b) (iii)	3	9.5.2.2.4, 14.1a,d	H9, H14
28 (c) (i)	3	9.5.3.3.2, 9.5.3.2.2, 13.1, 14.1	H13, H14
28 (c) (ii)	3	9.5.3.3.2, 9.5.3.2.7, 13.1, 14.1	H13, H14
28 (d)	6	9.5.4.1, 9.5.1.2.1, 9.5.3, 14.3b	H10, H14

Section II

Question 29 — Medical Physics

29 (a) (i)	2	9.6.2.3.1, 9.6.2.2.1	H10
29 (a) (ii)	4	9.6.2.2.1, 13.1e	H10, H13
29 (b) (i)	1	9.6.1.3.5	H8, H12, H13
29 (b) (ii)	1	9.6.1.2.6	H8
29 (b) (iii)	4	9.6.1.2.2	H8
29 (c) (i)	4	9.6.2.2.4, 9.6.2.2.3	H10
29 (c) (ii)	3	9.6.3.2.2., 9.6.3.2.3, 9.6.3.2.4, 12.3e	H10, H12
29 (d)	6	9.6.4.2.4, 9.6.4.2.5, 9.6.4.2.6, 9.6.4.3.3, 13.1a, 14.3b	H8, H9, H13, H14

Section II

Question 30 — Astrophysics

30 (a) (i)	2	9.7.1.2.3	H10
30 (a) (ii)	4	9.7.1.2.2, 9.7.1.2.4, 9.7.1.2.5, 9.7.1.2.3	H8, H10
30 (b) (i)	2	9.7.5.3.1	H8, H10
30 (b) (ii)	2	9.7.5.3.2, 12.4b	H12, H13
30 (b) (iii)	3	9.7.3.2.4, 9.7.3.2.5, 9.7.5.2.1	H14
30 (c) (i)	1	9.7.4.2.1, 9.7.4.2.4	H13, H14
30 (c) (ii)	2	9.7.4.2.4	H13, H14
30 (d)	3	9.7.4.2.5	H2
30 (e)	6	9.7.6.2.1, 9.7.6.2.3, 9.7.6.3.2, 14.3b	H14

Question	Marks	Content	Syllabus outcomes
Section II			
Question 31 — From Quanta to Quarks			
31 (a) (i)	2	9.8.1.2.1	H2
31 (a) (ii)	4	9.8.1.2.2	H10
31 (b) (i)	2	9.8.2.2.1	H8, H10
31 (b) (ii)	3	9.8.2.2.3	H8, H10
31 (b) (iii)	2	9.8.2.3.1, 12.4b	H8, H10, H12
31 (c) (i)	1	9.8.3.2.9	H7, H8, H10, H12, H14
31 (c) (ii)	2	9.8.3.2.8, 12.3c, 14.1a	H8, H10, H12, H14
31 (d)	3	9.8.3.2.2	H8, H10
31 (e)	6	9.8.4.2.5, 14.3b	H1, H8, H10, H14
Section II			
Question 32 — The Age of Silicon			
32 (a) (i)	1	9.9.2.2.4, 9.9.2.2.5	H13, H14
32 (a) (ii)	1	9.9.2.3.3, 12.4b	H12, H13
32 (a) (iii)	2	9.9.6.2.5, 9.9.6.3.5	H9, H14
32 (a) (iv)	2	9.9.5.2.1, 9.9.5.3.1	H9, H14
32 (b) (i)	2	9.9.3, 9.9.4	H9, H14
32 (b) (ii)	5	9.9.2.2.6, 9.9.3.2.2, 9.9.3.2.4, 9.9.3.3.1, 9.9.4.2.1, 9.9.4.3.1,	H7
32 (c)	2	9.9.2.3.1, 9.9.2.3.2, 9.9.2.2.3	H7
32 (d)	4	9.9.1.3.1, 9.9.1.2.2, 9.9.3.1, 9.4.3.2.8	H4
32 (e)	6	9.9.1.2.3, 9.9.1.3.1, 9.9.2.2.3, 9.9.2.3.2, 9.9.6.3.5, 9.9.7.2.1, 9.9.7.3.1, 14.3b	H3, H14